JONGHO PARK

Software Engineer

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Professional Profile

Software developer at Zetaris with commercial experience in React/TypeScript, Java/Spring REST APIs and Bring Your Own Cloud (BYOC) deployments across AWS, Azure and OCI. Builds Azure-based test systems and Azure DevOps CI pipelines to support customer-specific validation and reliable releases. Independently developed RAG and custom agentic LLM workflows using OpenAI APIs, prompt and context engineering, and retrieval evaluation. Holds a Master of Data Science from the University of Melbourne.

Technical Skills

Software engineering: Python, TypeScript/JavaScript, Java, SQL; React, Next.js, Spring Boot, REST APIs

Generative and agentic AI: LLMs, OpenAI API, RAG, prompt and context engineering, custom agent workflows, retrieval evaluation

Cloud platforms: AWS, Microsoft Azure (Container Registry), Oracle Cloud Infrastructure (OCI), GCP (BigQuery), Cloudflare Workers

Testing and delivery: Azure-based test systems, Azure DevOps CI/CD, Playwright, JUnit, Allure, Docker; Git, Claude Code

Reliability and security: background job processing, retries/DLQ, generation metrics, secret redaction and audit logging

Professional Experience

Junior Software Developer — Jul 2026 - Present

Zetaris

- Operated and validated Bring Your Own Cloud (BYOC) deployments across AWS, Azure and OCI, tailoring test scenarios to customer environments and contributing bug fixes to support reliable releases.
- Built and enhanced Azure-based test systems and Azure DevOps CI pipelines, integrating backend-connected Playwright tests and Allure reporting to verify data persistence, permissions and release behaviour.
- Built Java/Spring Boot APIs and a React admin interface to control local deployment access and revoke Azure Container Registry credentials, with transaction locking, background retries and audit logging.
- Built merchant-risk scenarios and validation assets for an AI Data Steward demo for prospective customers, including Python-generated synthetic datasets, SQL reference answers, semantic models, graphs and supporting documents.
- Improved React/TypeScript data exploration and ingestion workflows through reusable table and form components, folder selection validation and predictable catalog navigation.
- Hardened deployment diagnostics with content-based secret redaction across Helm, Kubernetes and application logs; added unit and integration regression tests for failure paths.
- Built an Electron/DuckDB Lakehouse Sync prototype with resumable multipart uploads, change tracking and repair; validated a 14.8 GB Parquet load of 99,997,497 rows against a hosted test environment.

Data Science Intern — Mar 2023 - Nov 2023

Neighbourlytics, Melbourne

- Built BigQuery geospatial processing workflows over approximately 2 million mobile-device pings, preparing data for spatial analysis and behavioural segmentation.
- Developed preprocessing and validation workflows, evaluated DBSCAN and spectral clustering, and documented schemas to support reproducible analysis.

Selected Engineering Projects

ResearchQ AI Research Publishing Platform

Independent project | React, TypeScript, OpenAI API, Cloudflare Workers, D1, Vectorize, Queues

- Built a multilingual RAG pipeline and evaluation harness; diagnosed an embedding-language mismatch and improved self-retrieval Recall@5 from 0.05 to 0.95 on 20 title-based queries.
- Implemented custom agentic workflows with OpenAI APIs for planning, context retrieval, drafting and self-review, using structured audience instructions and human-controlled publication.
- Built and deployed a React/TypeScript application and Cloudflare Workers backend with 75 API endpoints, relational data storage and vector search.
- Implemented background job processing for generation, indexing and evaluation using Cloudflare Queues, typed messages, automatic retries and a dead-letter queue.
- Recorded per-stage latency and output metrics, persisted retrieval evaluation results and captured failed drafts to support diagnosis and comparison between runs.
- Refactored a 4,494-line React component into 13 focused modules, reducing the main file to 2,651 lines through 11 build-verified increments.

Cloud Social Sentiment Analytics Platform

University of Melbourne team project | Python, Docker, Ansible, OpenStack, CouchDB

- Contributed to a distributed cloud platform for ingesting and analysing Twitter and Mastodon posts, combining NLP workflows with time-series and geographic analysis.
- Used Docker and Ansible for repeatable multi-node deployment on Melbourne Research Cloud, with CouchDB storage and MapReduce views for distributed aggregation.

Education

Master of Data Science 2023 - 2025

The University of Melbourne

Machine learning, statistical modelling, cloud and distributed computing.

Bachelor of Science in Data Science 2020 - 2022

The University of Melbourne

Foundations in algorithms, probability, statistics and database systems.

Additional Experience

Signals Intelligence Sergeant Jan 2024 - Jul 2025

Republic of Korea Army

- Led a team of 10+ personnel and standardised intelligence reporting workflows, maintaining data quality and consistency in time-sensitive operations.

Certifications

AWS Cloud Practitioner | ADsP (Data Analytics Semi-Professional)